

**SECTION B**

*Answer all questions in the spaces provided.*

6. (a) The atomic spectrum of hydrogen consists of several separate series of lines.
- (i) In the Balmer series, when an electron returns from the third shell to the second shell, a red line is seen at a wavelength of 656 nm. Calculate the energy of the radiation emitted at this wavelength. [3]

Energy = ..... J

- (ii) Explain how the Lyman series can be used to calculate the ionisation energy of hydrogen. [2]

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- (b) State and explain how you would expect the ionisation energy of hydrogen to compare with the first ionisation energy of:

- (i) helium [2]

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- (ii) lithium [2]

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- (c) (i) On adding a strip of magnesium to hydrochloric acid,  $125\text{ cm}^3$  of hydrogen gas formed at a temperature of  $24^\circ\text{C}$  and a pressure of  $1.01 \times 10^5\text{ Pa}$ . Calculate the volume occupied by this gas under the same pressure and at a temperature of  $52^\circ\text{C}$ . [2]

Volume = .....  $\text{cm}^3$

- (ii) In another experiment,  $160\text{ cm}^3$  of hydrogen formed at  $20^\circ\text{C}$  and  $1.01 \times 10^5\text{ Pa}$ . Assuming that hydrogen behaves as an ideal gas, calculate the amount, in moles, of hydrogen formed. [3]

n = ..... mol



- (d) Hydrogen can form a range of covalent hydrides such as water,  $\text{H}_2\text{O}$ , and beryllium hydride,  $\text{BeH}_2$ . A student said that since the ratio of hydrogen to the other element is the same in both compounds, the shapes of the two compounds will be the same. Is she correct? Justify your answer. [3]

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- (e) (i) On forming ice at  $0^\circ\text{C}$ ,  $50.0\text{ cm}^3$  of water expands to occupy  $54.5\text{ cm}^3$ . Calculate the density of ice at  $0^\circ\text{C}$ . [2]

Density = .....  $\text{g cm}^{-3}$

- (ii) Explain why ice and water have different densities at  $0^\circ\text{C}$ . [1]

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